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Recycling of a post-industrial printed multilayer plastic film containing polyurethane inks by solvent-targeted recovery and precipitation

Kevin L. Sánchez-Rivera^a, Aurora del Carmen Munguía-López^a, Panzheng Zhou^a, Victor S. Cecon^b, Jiuling Yu^a, Kevin Nelson^c, Daniel Miller^c, Steve Grey^c, Zhuo Xu^{a,d}, Ezra Bar-Ziv^d, Keith L. Vorst^b, Greg W. Curtzwiler^b, Reid C. Van Lehn^a, Victor M. Zavala^a, George W. Huber^{a,*}

^a Department of Chemical and Biological Engineering, University of Wisconsin-Madison, Madison, WI, 53706, United States of America

^b Food Science and Technology, Polymer and Food Protection Consortium, Iowa State University, Ames, IA, 50011, United States of America

^c Amcor, Neenah Innovation Center, Neenah, WI, 54956, United States of America

^d Department of Mechanical Engineering, Michigan Technological University, Houghton, MI, 49931, United States of America

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ABSTRACT

The recently reported Solvent-Targeted Recovery And Precipitation (STRAP) strategy is used to deconstruct multilayer plastic packaging films into their constituent resins by selective polymer dissolution. To broaden the applicability of STRAP, we demonstrate the process with a flexible post-industrial printed multilayer plastic film used for food packaging. The material was a reverse printed film composed of polyethylene (PE), ethylene vinyl alcohol (EVOH), polyethylene terephthalate (PET), and polyurethane (PU)-based inks. Removal of all colors was observed with biomass-derived gamma-valerolactone (GVL) and the recovered polymers possessed comparable properties to the corresponding virgin resins. A technoeconomic analysis indicates that this STRAP process could be economically feasible at a processing capacity of 6,000 tons per year. Moreover, the production of plastic films with materials recovered by this STRAP process can have a lower climate change impact than the production of films from virgin polymers.

1. Introduction

Most of the plastic waste generated globally from 1950 to 2015 has been discarded and is accumulating in landfills or in the environment (Geyer et al., 2017; Zhao et al., 2018). Only 9% of this plastic waste has been recycled. Through the improvement of existing recycling technologies and the development of new ones, recycling rates of different plastic materials could be increased, including plastic packaging which is one of the largest sectors for single-use plastics (Geyer et al., 2017; Giacobelli et al., 2018). In the plastic packaging industry materials can be manufactured as monolayer or multilayer packaging articles. Typically, monolayer plastic films are used in stretch film applications and in agriculture and construction (Horodytska et al., 2018). On the other hand, multilayer plastic films are composed of different polymeric or non-polymeric materials (like aluminum or paper) and these are combined to achieve desired properties for the overall packaging (Coles et al., 2011). Typical polymers found in multilayer films include

polyethylene (PE), polypropylene (PP), polyethylene terephthalate (PET), polystyrene (PS and high impact polystyrene, HIPS), polyvinyl chloride (PVC), polyvinylidene chloride (PVDC), polyamides (PA) or nylons, ethylene vinyl alcohol (EVOH), ethylene-vinyl acetate (EVA) and other materials like papers, foils, inks, and additives (Coles et al., 2011; Kaiser et al., 2018). Each layer contributes unique attributes to the final properties of the film. For example, HDPE provides a moisture barrier, EVA provides sealability, polyamides provide mechanical integrity, and EVOH layers provide oxygen barrier properties (Niaounakis, 2020).

Flexible packaging has benefits over other types of packaging, like less required material and energy for production than rigid containers, lighter weight for transportation, shelf-life extension of food products, visibility of contents, and the capability of providing necessary product information (Niaounakis, 2020; F.P.A. (FPA) 2023). In most cases, these materials are intended for single use and traditional recycling technologies like mechanical recycling cannot be used for their separation

* Corresponding author.

E-mail address: gwhuber@wisc.edu (G.W. Huber).

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(Niaounakis, 2020; Garcia and Robertson, 2017). Current recycling rates of flexible packaging have been around 1 – 2% in the United States, which has led to these being incinerated or landfilled (Niaounakis, 2020; Bedard et al., 2021). In a recent report, it was estimated that the global flexible packaging market was at 31.5 million tons in 2021 and it is expected for this market to grow annually at a rate of 3.4% to 37.5 million tons in 2026 (Cameron, 2021). Flexible plastic packaging has many benefits, and are necessary to meet the needs of different sectors. Further technology development is required for more efficient collection, sorting, separation, and recycling of these flexible packaging materials to reduce the amount that end up in landfills or the environment, considering their expected production growth.

A method for recycling multilayer plastic films is the deconstruction of these materials via selective dissolution to separate their constituent polymers. This method has been studied by a number of researchers and it is currently being implemented by several companies (Niaounakis, 2020; Accelerating circular supply chains for plastics 2019). We have demonstrated this approach in what we call Solvent-Targeted Recovery And Precipitation (STRAP) (Walker et al., 2020; Sánchez-Rivera et al., 2021). The STRAP process has been used to convert a commercially available post-industrial rigid multilayer film manufactured by Amcor into its three main polymer components with >99 wt% recovery. This multilayer film consisted primarily of polyethylene (PE), ethylene vinyl alcohol (EVOH) and polyethylene terephthalate (PET) with various tie layers that included ethylene vinyl acetate (EVA). In this process, the film is dissolved in two different solvents, each of which solubilizes an individual polymer. The final resin is then precipitated with the addition of an antisolvent or by cooling the solvent system. The solvent selection for STRAP is based on quantifying polymer-solvent thermodynamic properties using three computational methods of increasing accuracy and complexity: Hansen Solubility Parameters (HSPs), molecular dynamics (MD) simulations, and a combined quantum chemical and statistical mechanical approach called the CONductor-like Screening MOdel for Realistic Solvents (COSMO-RS) (Zhou et al., 2021). In general terms, HSPs are used for the preliminary solvent screening for the selective dissolution of specific polymers, MD simulations provide structural information of polymer molecules in solution, and COSMO-RS enables predictions of polymer solubility as a function of both the temperature and the composition of the liquid phase (Barton, 1975; Klamt et al., 2010; Eckert and Klamt, 2002; Belmares et al., 2004; Díaz et al., 2013; Hansen, 2007).

Delamination is another approach for the recycling of multilayer plastic films. In this approach, the adhesive between polymer layers is selectively dissolved or reacted to achieve a separation of the different layers. Various media have been utilized for the delamination of multilayer plastic materials. For example, O'Rourke, et al. used glycolysis to target polyurethane (PU) adhesives for the separation of polyamide and polyolefin layers from a multilayer film (O'Rourke et al., 2022). In another example, Ügdüler et al. demonstrated the use of carboxylic acids for the delamination of PU adhesives in different multilayer plastic films. In that study, the authors determined that temperature and solvent to liquid ratios had significant effects on the delamination rates and fast delamination kinetics can be very beneficial in terms of potential reduction of CO₂ emissions when compared to incineration (Ügdüler et al., 2022). Berkane et al. studied the delamination of aluminum (Al) interlayers with and without performing microperforations to the multilayer material. They achieved the dissolution of the Al with a 10% sodium hydroxide (NaOH) solution (Berkane et al., 2023). Samorì, et al. developed a novel procedure with switchable hydrophilicity solvents to recycle multilayer materials containing PE and Al (Samorì et al., 2017). In their work, the plastic and metal components were recovered by adjusting the polarity of N, N-dimethylcyclohexylamine (DMCHA) via CO₂ addition. Other studies have been done with mixed textile waste where elastane can be recovered via selective dissolution with solvents like tetrahydrofurfuryl alcohol (THFA), gamma-valerolactone (GVL), cyrene,

dimethylacetamide (DMCA), n-methylpyrrolidone (NMP) or dimethylformamide (DMF), depending on the other materials present in the waste stream (Phan et al., 2023). Polymers recovered by the dissolution-precipitation method can be repurposed for alternative applications. For example, Cavalcante, Hardian, and Szekely recently demonstrated the dissolution of PP from disposable face masks using a green solvent, p-cymene. In their study, the recovered PP was used for the fabrication of membranes for organic solvent nanofiltration (Cavalcante et al., 2022). Other green solvents like α -pinene and D-limonene were demonstrated for the dissolution of polyolefins that can be used in similar membrane applications (Ramírez-Martínez et al., 2023). All these recent academic efforts demonstrate the interest in solvent-based recycling technologies and their potential for the recovery of various components commonly found in different plastic waste streams.

Flexible packaging films are printed to help market the product and provide product information (Coles et al., 2011). There is significant interest in the removal of colorants from plastics since these are considered impurities that can lead to colored plastic streams which can limit applications when being recycled (Billiet and Trenor, 2020). Color removal can be a difficult task since inks can be water-borne or solvent-borne and their compositions vary with each color and ink manufacturer. Inks can contain insoluble organic and inorganic pigments, binders, solvents, and additives like dispersing agents, stabilizers, slip agents, biocides, optical brighteners, among others (Sutter et al., 2011). These inks can be printed on polymeric substrates via different methods, one of them being reverse printing, which is a method where the printed ink is located as a trapped layer within the packaging material. This provides ink protection and avoids direct product or human contact (Durling, 2009; Kaiser et al., 2018; Somal, 2019; Dixon, 2011). On the contrary, in surface printing, the printed ink is not buried in between layers of packaging materials and can be optionally coated with an overprint or lacquer (Durling, 2009). For reverse printed films, color removal is more challenging since the ink is buried between different plastic layers.

In this paper, STRAP was applied to a printed flexible multilayer plastic film to show the potential of this approach to recover and produce clear polymer resins. This was demonstrated with a post-industrial oriented polyester film (OPET) that comprises PE, EVOH, and PET, with polyurethane (PU) inks (Fig. 1). A challenge with this specific plastic waste feedstock is how and at what stage of the process should the ink components be removed. Different surfactant solutions and PU-selective solvents were considered to achieve the recovery of the polymer fractions without coloration.

2. Materials and methods

2.1. Computational modeling

Following our previous work, we used a combination of Hansen Solubility Parameters (HSPs) and the Conductor-like Screening Model for Real Solvents (COSMO-RS) to identify solvents for STRAP (Walker et al., 2020; Sánchez-Rivera et al., 2021; Zhou et al., 2021). HSPs are three parameters associated with each compound (polymer or solvent) that account for dispersion, polar and hydrogen-bonding forces (Hansen, 2007). The values of these three parameters define the coordinates of a compound in HSP space. Each polymer has an additional radius parameter, R_0 , that defines a sphere in HSP space, as shown in Fig. 2a. Only solvents with HSPs within this sphere (such that the distance in HSP space, R_d , between the polymer and solvent HSP values is less than R_0) are expected to dissolve the polymer. Tabulated, experimentally determined HSP values for polymers and solvents were obtained from literature sources (Hansen, 2007; Rasool and Vankalecom, 2021; Abbott and Hansen, 2008). COSMO-RS uses statistical thermodynamic methods to compute the equilibrium properties of multicomponent systems based on the screening charge density that arises at each compound's molecular surface due to the polarization of the medium (Klamt, 1995).

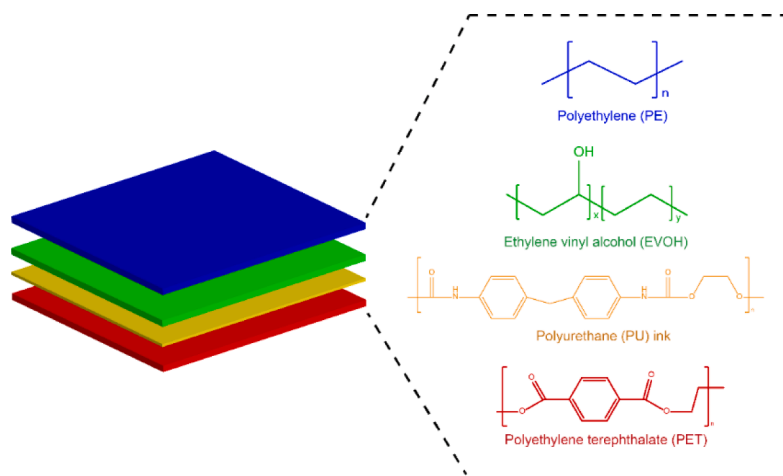


Fig. 1. Simplified structure of an OPET multilayer printed film composed of PE, EVOH, PET, and PU-based inks, manufactured by Amcor.

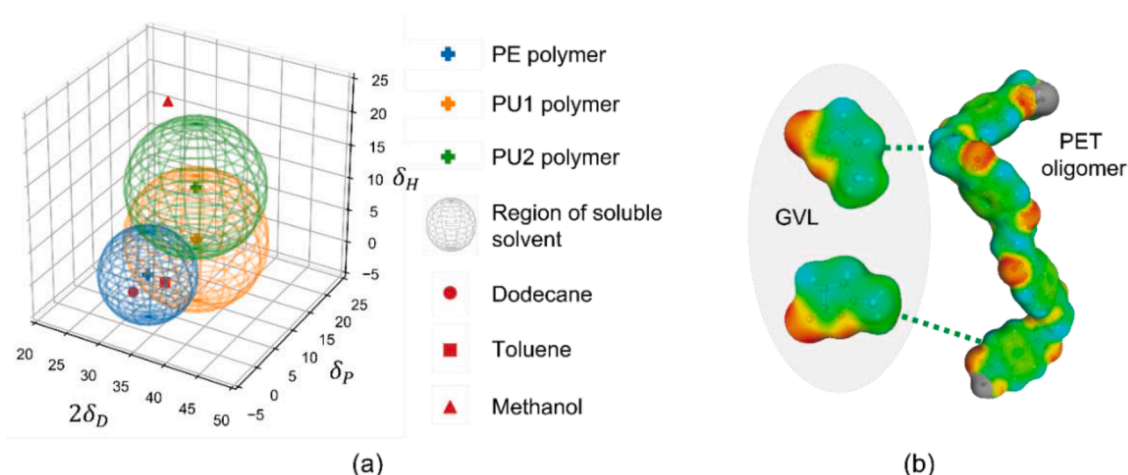


Fig. 2. Computational methods used in this work. (a) Screening of PE-selective solvents in the presence of PU-based inks using HSPs. The HSPs for the PE polymer, two representative PU polymers, and 3 example solvents are indicated as points in HSP space, with the soluble regions for each polymer drawn as spheres centered on the corresponding polymer HSPs. Dodecane lies within the PE sphere but outside of the PU spheres and is thus predicted to be a selective solvent for PE. (b) COSMO-RS screening charge distributions (colored surfaces) of GVL and a PET oligomer. Oligomer molecules with end groups neglected (gray surface) are used to represent the chemical properties of longer polymer chains. The intermolecular interactions between polymers and solvents are quantified by the interaction of screening charges, as schematically indicated by dashed lines.

Screening charge density profiles were obtained from density functional theory (DFT) calculations performed using Gaussian 16 at the BVP86/TZVP/DGA1 level of theory (Frisch et al., 2016; Palomar et al., 2007; Liu et al., 2016). Screening charge densities were then used to predict polymer solubilities via a solid-liquid equilibrium calculation using the COSMOtherm 19 software with the BP_TZVP_19 parameterization (Zhou et al., 2021; Eckert and Klamt, 2002; Klamt et al., 2002). This solubility calculation requires the polymer melting temperature and an experimentally measured solubility in a reference solvent as input, which were obtained using data from our previous work (Zhou et al., 2021; Zhou et al., 2023).

2.2. Materials

The printed flexible multilayer film was collected from an Amcor post-industrial waste stream. The materials were initially shredded through a 1/8" cross-cut shredder (Make: Allegheny 16-75CX) at Michigan Technological University. The printed film was further shredded with a Thomas Model 4 Wiley Mill to 8, 4, 3, and 1 mm size for the dissolution tests with different sizes. The pure virgin resins, LLDPE, LDPE, EVOH, and PET, were received from Amcor. Dodecane

(ReagentPlus®, ≥99%), GVL (BioRenewable, ≥99%, ReagentPlus®), dimethylsulfoxide (DMSO, ACS reagent, ≥ 99.9%), and hexadecyltrimethylammonium bromide (CTAB, ≥ 98%), were received from Sigma-Aldrich.

2.3. STRAP experimental procedure

A 250 mL round bottom flask connected to a reflux condenser with a cold-water supply line was used for the STRAP experiments. The round bottom flask, which would contain the corresponding solvent and shredded multilayer film with a ratio of 10:1 (by mass), was partially submerged in a 1500 mL dish containing silicone oil as a heat transfer fluid. The system was heated to the desired dissolution temperature with an electric heat plate equipped with a magnetic stir drive and the stirring rate was adjusted to have constant mixing. Based on the computational analysis for solvent selection, we experimentally assessed the recovery of each polymer in the printed multilayer film by performing three processing steps: (1) selectively dissolving the PE fraction in dodecane at 95 °C for 30 min, then separating the solubilized fraction from the EVOH and PET via mechanical filtration; (2) selectively dissolving the EVOH fraction in a 60% DMSO-40% water (v/v) mixture at 95 °C for 60 min,

then separating the solubilized fraction from the remaining PET and ink via mechanical filtration; and (3) removing ink from PET using GVL at 125 °C for 60 min. The PE and EVOH fractions were precipitated by reducing the respective solvent temperatures to 35 °C and then the solvents were separated via filtration. Similarly, the PET was separated from the GVL and ink via mechanical filtration. The resulting PE and EVOH solid fractions were dried in a vacuum oven for 3 h at 100 °C and −1 bar to remove residual solvents. The PET fraction was dried for 3 h at 150 °C and −1 bar. Sequential PE-dissolution steps were required to recover most of the polymer and clean out the experimental system. For the deinking step, color removal improved with sequential steps under the same conditions, longer deinking times, and/or higher temperatures.

2.4. Characterization

Samples obtained from STRAP were analyzed with ATR-FTIR (Bruker Vertex 70) to verify polymer purity. PE samples were analyzed with high-temperature GPC (Malvern Viscotek 350 HT-GPC) to obtain molecular weight information. Thermal properties of the samples were analyzed with DSC (TA Instruments Q2000 differential scanning calorimeter) and TGA (TA Instruments Q5000IR thermogravimetric analyzer). Final color of the recovered polymers was quantified with a spectrophotometer and measurements were converted to CIELAB color space values, L^* (lightness), a^* (red-green color component) and b^* (blue-yellow color component), and YI (yellowness index) (ASTM D6290-19 2019). For detailed characterization information, see the Supplementary Information.

2.5. Technoeconomic analysis (TEA) and life cycle assessment (LCA) methodology

The process design and TEA were performed in the open-source platform BioSTEAM (Cortes-Peña et al., 2020). This Python-based process simulator has been validated against proprietary software (Super-Pro Designer and Aspen Plus). The collected experimental data was used to simulate the STRAP process in BioSTEAM. Here, mass and energy balances are used to determine the sizing and the energy required by the different process units (e.g., heat exchangers and distillation). The impact of the STRAP process with the printed multilayer film on climate change (kg CO₂ eq.) was evaluated using a life cycle assessment (LCA) methodology. The STRAP process was analyzed from a product perspective (Jeswani et al., 2021). From this perspective, the STRAP technology is seen as an alternative process to produce virgin-grade multilayer films. The functional unit considered is the production of 1 kg of multilayer film. Thus, we compare the impacts of producing 1 kg of film from traditional pathways with producing 1 kg of film through the STRAP process. The LCA was performed using the open-source software openLCA v1.10.3 (Ciroth et al., 2020). The solvent impacts were taken from the Ecoinvent 3.6 cut-off by classification database (Wernet et al., 2016) using the Environmental Footprint impact assessment method (Zampori and Pant, 2019). The rest of the impacts were taken from the Environmental Footprint database (Fazio et al., 2018).

3. Results and discussion

3.1. Ink removal from the printed multilayer film

Non-ionic, cationic, and anionic surfactant solutions have been used for efficient ink removal from plastic substrates (Gecol et al., 2003; Gecol et al., 2002). One common cationic surfactant is hexadecyltrimethylammonium bromide (CTAB) and solutions of this surfactant were prepared at different concentrations (0.1–10 wt%) for deinking of the printed multilayer film at different temperatures and pH. The solutions were above the critical micelle concentration (CMC) of CTAB, which is 0.0009 M in water (Li et al., 2006). The printed multilayer film had yellow, white, and black as the main colors, with traces of blue and

red. The deinking step with CTAB was incorporated at the beginning of the STRAP experiments to target all colors, however, no ink removal was visible at various temperatures (25–65 °C), mixing rates (100–400 RPM), and pH (8 and 13) for the tested CTAB solutions in water. A pressurized system was also used to conduct the deinking with CTAB at 175 °C but no significant ink removal was observed. We believed this could be due to the surfactant not contacting the ink components in the multilayer plastic film structure since the film was manufactured using the reverse printing technique. Ink removal using surfactants consists of four steps: (1) absorption of surfactant on plastic surfaces, (2) solubilization of the ink binder in surfactant micelles, (3) release of ink particles from the surface, and (4) stabilization of the ink particles (Chotipong et al., 2007; Kol et al., 2021). Deinking may not occur if there is no contact between the surfactant and the printed surface, as it was observed with the tested printed multilayer film. For example, researchers at the Universidad de Alicante developed a technology in which the ink removal includes a step where the plastics are punctured. This was done for the deinking agent to reach the interlayer ink components (Font, 2021). The deinking step was also evaluated with CTAB after the first and second polymer dissolution steps but still no efficient ink removal was observed. A recently published study on different deinking treatments for printed plastics reported that CTAB solutions were not optimal for removing PU-based inks on monolayer plastic films, and high deinking efficiency of PU resin was observed with N, N-dimethylcyclohexylamine (DMCHA) or formic acid (Ügdüler et al., 2023). This gives further indication that CTAB solutions are not optimal for removing inks containing PU resins and other media need to be considered, along with potentially puncturing the plastics for deinking. In our study, since the selected CTAB surfactant solutions did not achieve any significant deinking, PU-selective solvents that could target the ink components were screened, using our computational modeling tools. After a deinking step, it would be preferred to have a physical ink/plastic separation so that both can be easily isolated. Dissolution-based ink separation could be more energy costly since the medium would have to be purified if a physical separation cannot be achieved, as it would be with some organic solvents. The selectivity for the polymer components in the film over PU was considered, building upon our experience with transparent multilayer plastic films.

3.2. Computational modeling results

In our previous work, we successfully recovered the main constituent polymers (PE, EVOH, and PET) from a transparent multilayer film through a STRAP process in which PE was first dissolved by toluene, EVOH was dissolved by a DMSO/water mixture, and PET was recovered as the only remaining resin (Sánchez-Rivera et al., 2021). The printed film considered in this work consists of the same three polymers with an additional PU-based ink. Therefore, we sought to modify the prior STRAP process by using computational tools to identify new solvents that remain selective to the target polymers in the presence of the ink, as no ink removal was observed with different surfactant solutions.

The first step is to find a selective solvent for the PE component of the printed film. Since the specific molecular structure of the PU-based ink is unknown, we cannot directly predict ink solubility with molecular-scale modeling. We first conducted quantitative solubility predictions using COSMO-RS for the three known polymers to identify potential selective solvents, then used qualitative HSP calculations to eliminate solvents likely to dissolve the PU-based ink. Table S1 in the Supplementary Information shows the predicted solubilities for PE, EVOH, and PET in 30 common solvents at elevated temperatures. We define PE-selective solvents as solvents with a predicted PE solubility greater than 10 wt% and predicted EVOH and PET solubilities that are each less than 5 wt%. The solubility values are defined as the mass of polymer than can be dissolved in a known amount of solvent at a given temperature (g polymer/g solution). Four PE-selective solvents were identified: toluene, dodecane, heptane, and diphenyl ether. Since information on the ink structure

is limited, we collected HSP data for 11 different PU resins to determine the potential of these four solvents to dissolve the ink. Table S2 presents HSP data for these PU resins and the 4 solvents studied and Table S3 reports calculated values of R_a/R_0 . Based on these data, toluene is predicted to dissolve ($R_a/R_0 < 1$) 4 of the 11 PU resins and diphenyl ether is predicted to dissolve 7, whereas dodecane and heptane are not predicted to dissolve any of the PU resins. Therefore, dodecane and heptane are possible computationally identified PE-selective solvents for the printed film. Dodecane was finally selected for experiments since it has a higher predicted PE solubility than heptane. The selectivity of this solvent was verified experimentally.

The second step is to find an EVOH-selective solvent after PE dissolution. In our previous STRAP process, we used a mixture of 60 % DMSO-40 % water (v/v) to selectively dissolve EVOH but not PET (Sánchez-Rivera et al., 2021). We thus considered whether the ink would dissolve in this solvent system. HSP calculations were again performed for this solvent mixture and the 11 PU resins with results shown in Table S3. None of the 11 PU polymers are expected to dissolve in this solvent mixture. Therefore, the same 60 % DMSO-40 % water (v/v) solvent system was used in experiments and selective EVOH dissolution was observed.

The final step is to remove the PU-based ink from the remaining PET resin after PE and EVOH separation. We performed HSP calculations to identify solvents that could dissolve PU but not PET. Since the HSP data

for the different PU resins varied substantially, we identified potential PU-selective solvents that are predicted to dissolve at least 5 of the 11 PU resins without dissolving PET. Table S4 shows HSP calculations for solvents that fit these criteria. Among these solvents, we noted gamma-valerolactone (GVL) as an appealing, bioderived and biodegradable green solvent with low toxicity (Kerkel et al., 2021). The R_a/R_0 value of GVL to PET (1.02) is very close to 1, which indicates that it may dissolve PET at elevated temperatures. We thus computed temperature-dependent PET solubilities in GVL using COSMO-RS, as shown in Figure S1. Using a 5 wt% solubility threshold to distinguish good and poor solvents, we predict that the temperature should be below 125 °C to avoid substantial dissolution of PET in GVL. Experiments have verified that PET can dissolve in GVL above 165 °C but not at 125 °C, which aligns with our computational predictions.

3.3. Experimental demonstration of STRAP with the printed multilayer film

The flexible printed multilayer film composed of PE, EVOH, PET, and PU-based inks was processed through STRAP to recover all polymer components. Fig. 3a shows solvents, temperatures, dissolution times, and polymer yields for each step. In our approach, the PE and EVOH are dissolved and recovered, leaving the PET and ink components behind. After this, the ink is removed from the PET component using GVL.

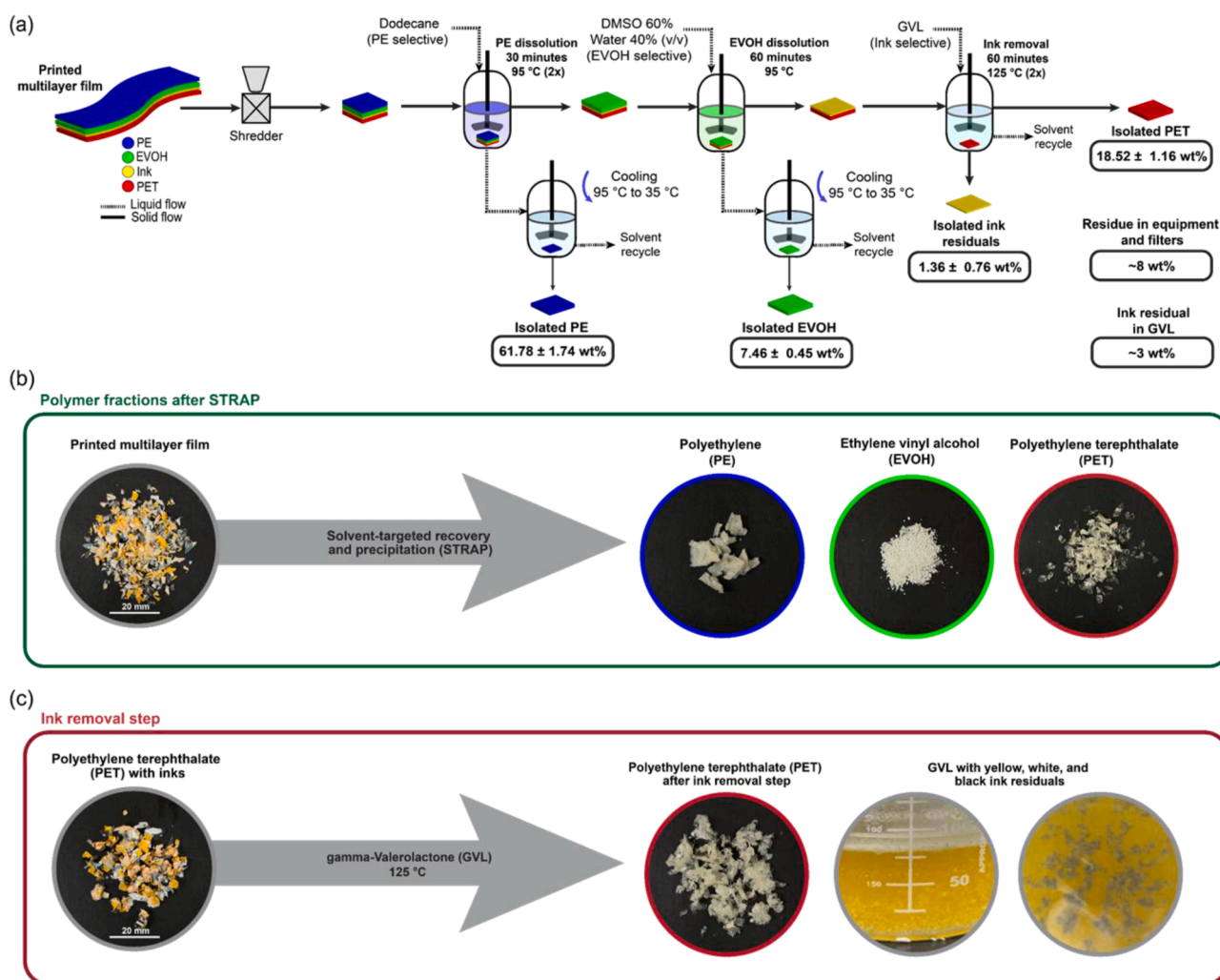


Fig. 3. (a) Solvent-targeted recovery and precipitation (STRAP) of a printed multilayer film composed of PE, EVOH, PET, and PU-based inks, (b) Photos of each polymer after the STRAP process, (c) STRAP deinking step with gamma-valerolactone (GVL) to separate white, black, and yellow ink from PET of a printed multilayer film.

Polymer precipitation was achieved by reducing the temperature of the respective solvents and not by the addition of antisolvents. In our previous work, this was demonstrated to be beneficial both economically and environmentally (Sánchez-Rivera et al., 2021). By visual inspection, the final recovered polymers after STRAP exhibited little to no coloration (Fig. 3b). Due to the high percentage of PE in the printed film, an additional PE dissolution step was required to recover most of the polymer and clean out the experimental setup. This additional PE dissolution step was done in every experiment for this analysis, which allowed for an average PE yield of 61.78 ± 1.74 wt%. The average overall mass balance from the STRAP experiments was 88.66 ± 0.53 wt %. We found that the material that was not recovered after STRAP was left behind in our equipment, mostly in the round bottom flasks and filters. The overall mass balance improved when accounting for the material lost in the equipment which was around 8 wt% of the plastic feed. Furthermore, part of the ink (mostly yellow color) can be recovered via distillation of the GVL, around 3 wt% (Table S8). The overall mass balance can be >95 wt% of the starting material, considering the average recovery of PE, EVOH, and PET. This gives indication that when dealing with flexible plastics that have been shredded, material will be left behind in the equipment which should be considered when thinking about larger scale systems. Furthermore, it was challenging for us to predict and measure consistent amounts of material lost in the equipment since it depends on multiple factors like plastic size, stirring rates, and it is subject to how the material is handled in the batch experiment.

The main colors in the PU inks were yellow, white, and black. PUs are commonly used as adhesives in multilayer packaging materials and as binders in printing inks for plastic substrates (Sutter et al., 2011; Dixon, 2011). The ink separation step uses GVL, which is a PU and PET-selective solvent, as it was determined by our computational predictions. The PET was not dissolved in the ink removal step since the temperature was 125 °C and PET has a measured experimental solubility of 0.025 wt% at this temperature. Under these conditions, deinking was observed for all colors (Fig. 3c). White and black ink components were dispersed in the GVL after the deinking time was completed, while the yellow component was dissolved. Deinking of the PET improved by modifying one of the following: adding another deinking step with new GVL under the same conditions, increasing the deinking time up to 2 h, or increasing the GVL temperature to 135 °C. The final color of each polymer was quantified with CIELAB measurements, and all color values are presented in Table 1. The PE and EVOH had comparable L* values (lightness) to the corresponding virgin resins and did not have significant a* (red-green component) and b* (blue-yellow component) values. The recovered PET samples had reduced L* values and higher b* values when compared to virgin PET. Out of the PET deinking conditions tested in Table 1, two sequential steps (or extractions) with GVL provided the lowest b* value. Extending the deinking time or incorporating additional sequential washes with GVL could further remove more of the yellow color from the PET. According to a recent study, PET swelling with GVL occurs at 120 °C (Chen et al., 2021) which could play a role in the release of white and black ink components since dissolution of those colors was not observed. The white and black pigment residues were recovered by filtration and the dissolved yellow ink residues were recovered by distillation of the GVL. In ink formulations, inorganic and/or organic pigments are used. For example, in white ink, titanium dioxide (TiO₂) pigment is commonly used. Researchers have disclosed the use of o-dichlorobenzene and o-xylene to remove TiO₂ from PE, but they were only able to remove 15% of the colorant (Vuorinen et al., 2016). Some of the solvents that have been used to target colors and pigments have been organic solvents and some have been chlorinated compounds, which raises concerns about toxicity, safe handling, and environmental impact (Ferreira et al., 2021; Garrigós et al., 2023). In this study, we demonstrated the use of a green solvent, GVL, to clean PET from black, white, and yellow PU-based ink; this by using a combination of dissolution and ink dispersion in the solvent.

Table 1

Color values for virgin resins and polymers recovered from a printed multilayer film by STRAP.

Name	Sample	L*	a*	b*	YI
LLDPE	1	93.98	1.66	−5.66	−9.51
	2	93.77	1.63	−5.56	−9.37
	3	93.81	1.71	−5.57	−9.32
EVOH	1	94.01	1.75	−5.81	−9.73
	2	93.77	1.78	−5.60	−9.33
	3	93.98	1.87	−5.86	−9.73
PET	1	93.87	1.79	−5.79	−9.68
	2	94.03	1.84	−5.79	−9.63
	3	93.52	1.73	−5.18	−8.59
STRAP PE	1	91.27	−0.10	−0.05	−0.17
	2	91.28	−0.29	0.45	0.65
	3	91.42	−0.39	0.61	0.88
STRAP EVOH	1	93.60	−0.52	−0.52	−1.40
	2	93.50	−0.53	−0.53	−1.42
	3	93.65	−0.26	−1.05	−2.20
STRAP PET1 (125 °C 1 h)	1	68.19	−2.09	21.68	54.68
	2	70.02	−2.43	22.78	55.71
	3	70.01	−1.94	20.37	50.06
STRAP PET2 (135 °C 1 h)	1	76.59	−1.42	17.04	38.47
	2	71.69	−0.99	19.38	47.37
	3	73.82	−1.80	19.97	46.65
STRAP PET3 (125 °C 2 h)	1	73.43	−1.17	17.82	42.26
	2	74.84	−1.84	16.58	37.89
	3	74.29	−1.55	17.04	39.53
STRAP PET4 (125 °C 3 h)	1	69.96	−0.53	18.41	46.52
	2	76.03	−1.03	15.12	34.61
	3	71.55	−0.60	15.66	38.54
STRAP PET5 (two extractions at 125 °C 1 h)	1	76.90	−0.36	12.94	29.78
	2	75.89	−0.12	13.42	31.52
	3	77.14	−0.08	13.80	31.93

L* = lightness value; black = 0; white = 100.

a* = red-green color value; a* > 0 is redder; a* < 0 is greener.

b* = yellow-blue color value; b* > 0 yellower; b* < 0 is bluer.

YI = Yellowness Index.

3.4. Effect of plastic size on dissolution yield

Previous STRAP experiments with transparent multilayer films were initially demonstrated with film sizes of 1 × 1 cm and this necessitated extensive dissolution times to recover the target polymers. We investigated the effects of film sizes over the dissolution time of the PE component in the printed multilayer film. This was tested in the PE-dissolution step using dodecane at 95 °C and 500 RPM. Initial steps with a dissolution time of 30 min yielded 61.78 ± 1.74 wt% of PE when the film sizes were > 3 mm. This yield was calculated from the amount of extracted PE over the amount of plastic feed in a single experiment. Additional size reduction of the printed multilayer films to 3 mm and 1 mm showed improvement in the PE dissolution time. As seen in Fig. 4, at a total PE dissolution time of 8 min, the PE recovery improves by reducing the size of the films to 3 mm. From 3 mm to 1 mm, a slight improvement in the PE yield was observed. The 8 min of dissolution time was divided in two steps, a 5-minute step and a 3-minute step. As explained in the experimental demonstration section, two steps were carried out to recover most of the dissolved PE due to its high concentration in the film, and this helped with cleaning out the experimental setup. From the above results, we concluded that by downsizing the plastic material to the proper size (1–3 mm) and having vigorous mixing and the required dissolution temperature, we were able to extract most of the PE with dodecane. In dissolution-based plastic recycling technologies, plastic sizes become important whenever short dissolution times are of interest.

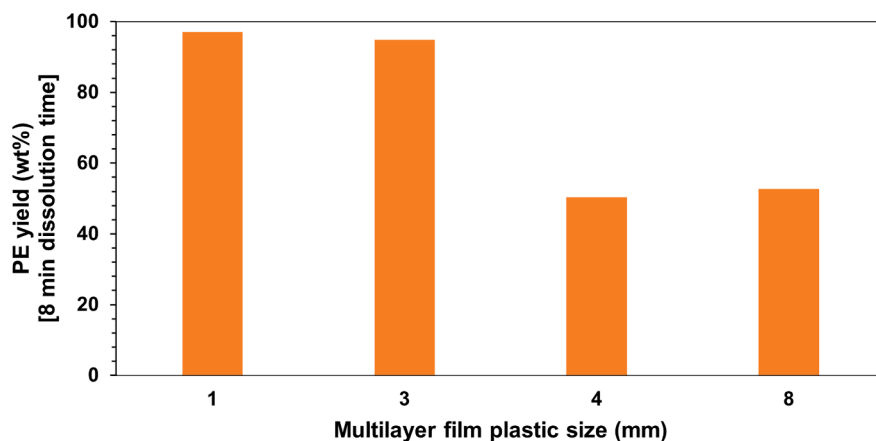


Fig. 4. Polyethylene (PE) recovery after 8 min of dissolution time with different plastic sizes. The recovery is based on the maximum PE yield obtained experimentally after 30-minute steps, around 61.78 wt%.

3.5. Characterization of polymers recovered by STRAP from the printed multilayer film

In the STRAP process with the printed multilayer film, the polymers can be recovered, and the ink components can be efficiently removed. In Fig. 5a, the IR spectra of the STRAP PE and PE virgin resin are compared, displaying identical spectral bands with no trace of any other bands belonging to other resins, indicative of the purity of the extracted PE. Fig. 5b and c show similar characteristics for EVOH and PET, respectively.

The molecular weight values of the recovered PE from the printed multilayer film by STRAP were determined with high-temperature gel permeation chromatography (HT-GPC). As seen in Figure S2, the number-average molecular weight (M_n), weight-average molecular weight (M_w), and dispersity (D) of the STRAP PE were within values of virgin LDPE and LLDPE resins. The printed film itself was manufactured with both LDPE and LLDPE, and the produced PE from STRAP had molecular weight values that were within the range of the pure resins. This gives evidence that the molecular weight of the PE is not significantly affected after being treated with dodecane in the STRAP process. The recovered PE, EVOH, and PET for the most part had similar thermal properties to

the corresponding pure virgin resins used in the starting material. For example, as seen in Table S10 in the SI, the melt temperatures ($T_{m,2}$) of the recovered polymers from the printed film by STRAP were comparable to the virgin resins. Similar results were obtained by Cecon, Curtzwiler and Vorst (Cecon et al., 2022) for the PE resins recovered using STRAP, with the thermal properties within the ranges displayed by the virgin HDPE and LLDPE resins used in a rigid transparent multilayer film. However, differences were observed in some parameters of the STRAP PET samples, possibly due to residual polymers or ink components. This could be addressed by carrying out additional ink removal steps to ensure PET is free of any contaminants. After STRAP, decreases in the crystallinities were observed for the recovered EVOH and PET, which could also be an indication of contamination in the samples. The previous characterization study of STRAP recovered materials did not determine significant changes in the crystallinity but in the melt temperature of PET, which was associated with solvent retention in the polymer matrix as PET was the main layer in the film (Cecon et al., 2022).

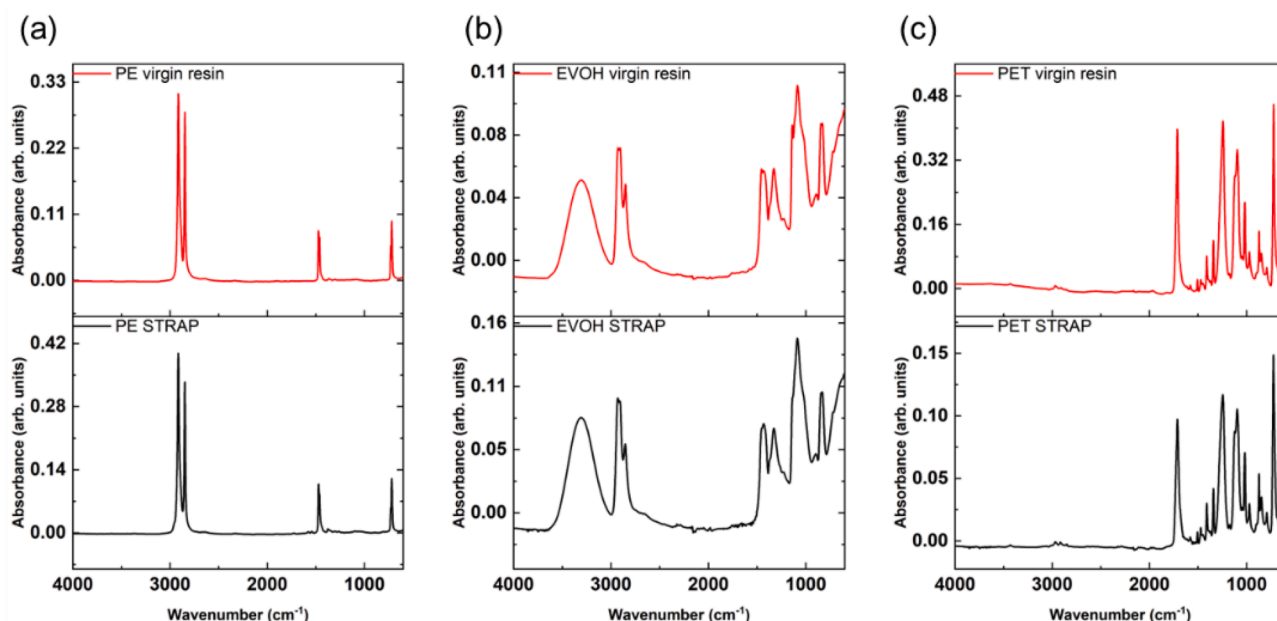


Fig. 5. ATR-FTIR spectra of virgin resins and polymers recovered by STRAP from the printed multilayer: a) PE, b) EVOH, and c) PET.

3.6. Economic and environmental impacts of STRAP with the printed multilayer film

The TEA and LCA of the STRAP process with the printed multilayer film were performed based on the collected experimental data. As described in the experimental demonstration section, part of the film could not be recovered and was left behind as residue in the equipment. Therefore, the material that could be recovered experimentally was normalized to close out the mass balance for the TEA and LCA. The sequential steps needed for PE and deinking of PET were considered in the amount of solvent and time required for those steps.

Figure S3 shows the process flow diagram used for the economic and environmental analyses with the main equipment units for the recovery of the different polymers: PE, EVOH, and PET. The TEA and LCA were performed considering three different scenarios presented in Figure S3: (a) only PE recovery, (b) PE and EVOH recovery, and (c) PE, EVOH, and PET recovery. This allowed for an estimate of the economics and environmental impacts of the STRAP process as a function of the number of polymer recovery steps. All the input (multilayer film and solvents) and output (recovered polymers and ink) streams are considered for the analyses. The steps of the process can be summarized as follows: (1) the printed multilayer film is shredded and then mixed with dodecane (1:20 ratio) in the dissolution vessel, (2) the plastic/dodecane is filtered to separate the dissolved PE from the non-dissolved plastics, (3) the PE is precipitated and the mixture is filtered again to separate the dodecane from the solid PE, (4) the liquid dodecane stream is recycled and the solid PE stream is dried to remove residual dodecane, and (5) the dodecane stream in the gas phase is condensed and recycled. Similarly, the EVOH separation can be summarized with the following steps: (1) the insoluble solids recovered after the first filtration step (EVOH, PET, and ink) are heated and mixed with a DMSO/water mixture in a dissolution vessel (1:10 ratio), (2) the DMSO/water/plastic mixture stream is then filtered to separate the dissolved EVOH from the non-dissolved PET and ink, (3) the stream containing the dissolved EVOH is cooled and the steps from the PE separation are repeated. Finally, the insoluble solids obtained after the EVOH separation, PET and ink, are heated and mixed with GVL in a dissolution vessel (1:20 ratio). After a first filtration, the PET is dried and the GVL stream in the gas phase is condensed and recycled. The remaining mixture of liquid GVL and ink is cooled and sent to a precipitation vessel. After a second filtration, the black and white ink residues are recovered. Then, a fraction of the remaining ink (yellow) with GVL (10%) is sent to distillation and the rest of the mixture is directly recycled along with the GVL recovered from distillation and the condenser. Our process simulation indicated that 99.98% of solvent recovery for GVL and 99.999% for the rest of the solvents (dodecane, DMSO, and water) is possible.

A summary of the economic and environmental impacts of the different scenarios considered for the STRAP process is shown in Fig. 6. The analysis was performed for a plant capacity of 3,000 tons per year, and the economic performance of each scenario is compared using the minimum selling price (MSP), which is expressed in USD per kg of

polymer sold. As expected, the PE recovery (around 62 wt% experimental yield) represented the lowest price, 1.96 USD/kg, and is the most attractive scenario economically because this value is within the average market values of the virgin and post-consumer recycled resins of 0.9–2.23 USD/kg (POLYMERSCAN, 2021). When the remaining polymers (EVOH and PET) are recovered, the price increases because the complexity of the process increases. However, in the detailed economic analysis (Section 3.6.1), it was found that at larger plant capacities the EVOH and PET recoveries can be economically feasible. The environmental costs in Fig. 6 were compared using the climate change impact (kg CO₂/kg of film) estimated using an LCA methodology. This metric refers to the generated emissions for the recovery of the different polymers via the STRAP process. The PE recovery also represented the lowest environmental impact, given that less consumption of resources is needed for one polymer recovery. In the detailed environmental analysis (Section 3.6.2), it was found that all the recovery scenarios have a lower impact than the alternative of producing films from virgin polymers.

3.6.1. Economies of scale for the STRAP scenarios with the printed multilayer film

In the economic analysis it was assumed that the economic lifetime of the plant is 20 years, the interest rate is 10%, and the input stream of multilayer film does not have any cost since it is post-industrial waste. The complete assumptions and parameters can be found in the Supplementary Information (Table S11). For the PE recovery step, it was determined that the total capital investment is 18.2 million USD and the operating costs are 1.24 million USD per year. The detailed capital and operating costs are reported in Tables S12–S16 of the Supplementary Information. The equipment that accounted for most of the capital cost were the extruder, dryers, and filters (Table S12). The operating costs include the steam and electricity required by each piece of equipment as well as the required solvent (dodecane), cooling agent, and natural gas. The solvent recovery in the process is 99.999%. A sensitivity analysis was conducted for different processing capacities to evaluate the impact of the economies of scale on the process of the PE recovery step. These results are presented in Figure S4a. We observe that after 3,000 tons per year, the MSP of recovered resins with the STRAP process is equal to average market values of the virgin resin and the price of post-consumer polymers (0.9–2.23 USD/kg) (POLYMERSCAN, 2021). These results demonstrate that the STRAP process is economically feasible at capacities larger than 3,000 tons per year. This occurs due to the effect of economies of scale, which allows for a reduction of the cost per unit by increasing efficiency. For the PE and EVOH recovery, at a processing capacity of 3,000 tons per year, we found that the MSP of the STRAP process is 2.4 USD/kg. This price is slightly higher than the average market values of the virgin resins and the price of post-consumer polymers (0.9–2.23 USD/kg) and higher than the MSP of just recovering the PE (POLYMERSCAN, 2021). The total capital investment is 26.4 million USD, and the operating costs are 1.6 million USD per year. The detailed capital and operating costs are reported in Tables S17–S21 in the SI. The solvent recovery in the process is 99.999% (for the three solvents used: dodecane, water, and DMSO). The economic sensitivity analysis for different processing capacities of the PE and EVOH recovery steps is presented in Figure S4b. We observe that after 4,000 tons per year, the MSP of the polymers recovered in the STRAP process is comparable to average market values of the virgin resins and the price of post-consumer polymers (POLYMERSCAN, 2021).

Finally, we analyzed the economic feasibility of the PE, EVOH, and PET recovery steps (including the ink removal). For the processing capacity of 3,000 tons per year, we found that the MSP of the STRAP process is 2.79 USD/kg. This price is slightly higher than the average market values of the virgin resins and the price of post-consumer polymers (0.9–2.23 USD/kg) (POLYMERSCAN, 2021) and higher than the MSP of just recovering the PE and EVOH. The total capital investment is 37.8 million USD, and the operating costs are 2.24 million USD per year.

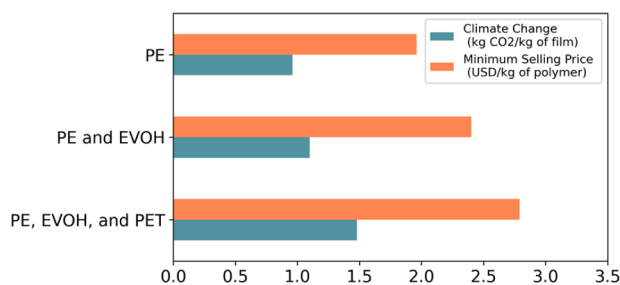


Fig. 6. Economic and environmental costs of the STRAP process with the printed film for different scenarios: PE recovery, PE and EVOH recovery, and PE, EVOH, and PET recovery (including ink removal).

The detailed capital and operating costs are reported in Tables S22–S26 in the Supplementary Information. The solvent recovery for the dodecane, water, and DMSO is 99.999%, while the recovery for GVL is 99.98% (because this recovery requires distillation). The sensitivity analysis for economies of scale is shown in Figure S4c. We observe that after 6,000 tons per year, the MSP of the resins produced in the STRAP process is comparable to average market values of the virgin resin and the price of post-consumer polymers (POLYMERSCAN, 2021). These results show that the separation of the 3 polymers is economically feasible at large processing capacities. There are important parameters of the process that can vary and impact the MSP. A sensitivity analysis was performed for selected parameters (OSBL, interest rate, and economic lifetime) and it was determined that the MSP is sensitive to the interest rate in all the scenarios, which can be found in Figure S5 in the SI.

3.6.2. Comparison of STRAP environmental impacts with multilayer film production from virgin resins

The impact of the STRAP process with the printed multilayer film on climate change (kg CO₂ eq.) was evaluated using a life cycle assessment (LCA) methodology. The STRAP technology was analyzed from a product perspective (Jeswani et al., 2021). From this perspective, the STRAP approach is seen as an alternative process to the production of multilayer films from virgin-grade polymers. The functional unit considered is the production of 1 kg of multilayer film. Thus, we compare the impacts of producing 1 kg of film from traditional pathways with producing 1 kg of film through the recovery of the polymers via the

STRAP process. The system boundaries of the production of multilayer film from fossil sources include the extraction and transportation of raw materials, material processing, and consumption of process utilities. In the STRAP process, the system boundaries include the material inputs (solvents and utilities), the transportation of the process solvents, and the material processing (extrusion) to produce the new multilayer films from recycled material. The film treated in the STRAP technology is assumed to be clean post-industrial waste. Therefore, the product use, waste collection, and treatment (e.g., sorting, grinding, and cleaning) are not within the system boundaries. The impacts for the solvents (DMSO and water) were taken from the Ecoinvent 3.6 cut-off by classification database (Wernet et al., 2016) using the Environmental Footprint method (Zampori and Pant, 2019). Since the impacts of dodecane and GVL were not available, we considered an average value of organic solvents from the same database. The rest of the impacts, including the STRAP process utilities (steam, electricity, and cooling agents), the transportation of commodities, and the virgin production of polymers (PE, EVOH, and PET) were taken from the Environmental Footprint database (Fazio et al., 2018). The environmental analysis section in the Supplementary Information presents additional LCA details, including assumptions, mass balances, inventory data, and a sensitivity analysis for the solvent recycling rates.

The environmental impacts of each of the three STRAP scenarios with the printed film were compared to the production of 1 kg of film with the corresponding virgin resins. As seen in Fig. 7a, the total impacts of separating PE by the STRAP process (0.96 kg CO₂ eq./kg film) are

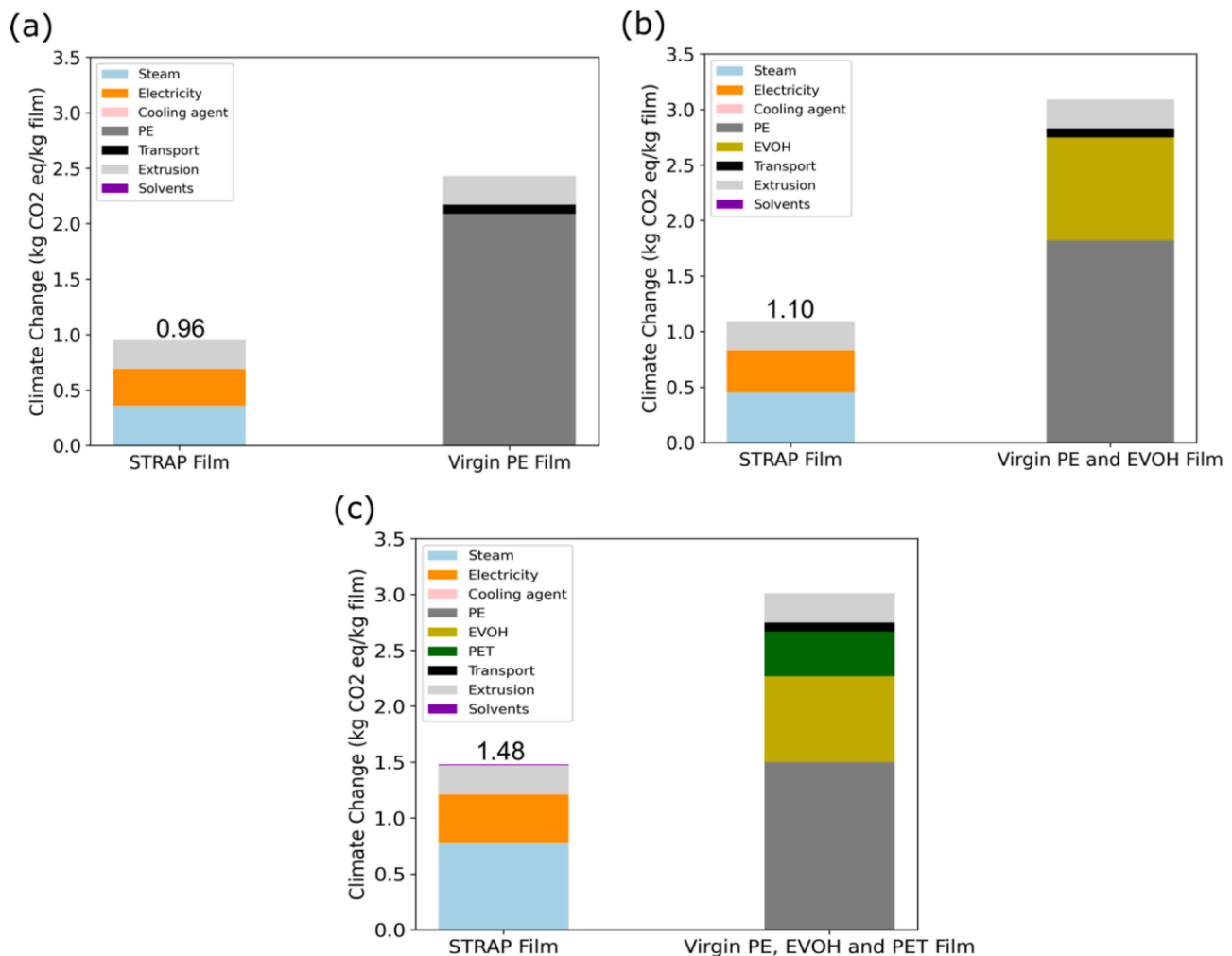


Fig. 7. Climate change impact of producing 1 kg of film with polymers recovered through the STRAP process in comparison with producing a film with virgin resins. a) PE film, b) PE and EVOH film, c) PE, EVOH, and PET film.

lower than the impacts of a PE film produced with virgin resins (2.43 kg CO₂ eq./kg film). Specifically, around 60% fewer emissions are generated in the STRAP technology. Most of the emissions in the STRAP process are related to the use of steam and electricity. Fig. 7b presents the generated emissions to produce a PE and EVOH film through the different alternatives. We can see how the impacts increase, but the STRAP approach keeps producing fewer emissions than virgin production despite including the impact related to solvents (DMSO and water). Finally, in Fig. 7c, the climate change impact of recovering all the constituent polymers (PE, EVOH, and PET) and producing 1 kg of virgin-grade multilayer film is presented. The generated emissions in the STRAP technology include the impacts related to the ink removal. The virgin film production generates around 51% more emissions (3 kg CO₂ eq./kg film) than the STRAP process (1.48 kg CO₂ eq./kg film). Through this analysis, we conclude that the STRAP technology could help achieve the goals of a circular economy by reducing the environmental impacts related to resource consumption and providing environmental benefits over virgin film production. These results are consistent with our previously published analysis (Munguía-López et al., 2023). Furthermore, the STRAP process has similar economic and environmental performance as delamination (O'Rourke et al., 2022).

4. Conclusions

The STRAP approach for recycling multilayer plastic films via selective polymer dissolution was applied to recover the constituent polymers of a printed flexible multilayer plastic film. A deinking step was included to remove all colors and obtain clear PE, EVOH, and PET that could be recycled for different applications. The computational modeling tools used in this study allowed for the selection of polymer-selective solvents in the presence of the inks and for the selection of solvents to develop an ink removal step. Out of the surfactant solutions and solvents tested, removal of all colors from the PET was observed with gamma-valerolactone (GVL), after recovering the other polymer components. This demonstrated the use of a biomass-derived solvent to clean PET from black, white, and yellow ink components. After quantifying colors with CIELAB measurements, the only polymer that showed residual colors was the recovered PET, particularly of the yellow color. Further improvement in the deinking step would be needed to reduce the amount of residual color in the PET stream. Other characterization results showed that, for the most part, the recovered polymers had comparable properties to the corresponding pure resins. Some differences were observed in crystallinity values which could indicate sample contamination from other polymers or residual ink. The separation of the polymers from the printed multilayer film with STRAP could be economically feasible at a large scale and it could reduce environmental impacts when compared to the production of films from virgin polymers. These results demonstrate that the STRAP framework, which combines computational modeling for solvent selection, experimental demonstrations, and process design, can be used to develop a viable recycling approach for a printed multilayer plastic film. Future work will focus on demonstrating this recycling approach with other multilayer films that contain PU inks, mixed multilayer plastic waste, and films that contain other polymers like PP, PS, and nylons.

CRedit authorship contribution statement

Kevin L. Sánchez-Rivera: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Aurora del Carmen Munguía-López:** Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Panzheng Zhou:** Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Victor S. Cecon:** Formal analysis, Investigation, Writing – review & editing. **Jiuling Yu:** Conceptualization, Investigation, Writing – review & editing. **Kevin Nelson:** Investigation, Writing – review & editing. **Daniel**

Miller: Investigation, Writing – review & editing. **Steve Grey:** Investigation, Writing – review & editing. **Zhuo Xu:** Investigation, Writing – review & editing. **Ezra Bar-Ziv:** Investigation, Writing – review & editing. **Keith L. Vorst:** Investigation, Writing – review & editing. **Greg W. Curtzwiler:** Investigation, Writing – review & editing. **Reid C. Van Lehn:** Investigation, Writing – review & editing. **Victor M. Zavala:** Investigation, Writing – review & editing. **George W. Huber:** Funding acquisition, Investigation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.resconrec.2023.107086](https://doi.org/10.1016/j.resconrec.2023.107086).

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